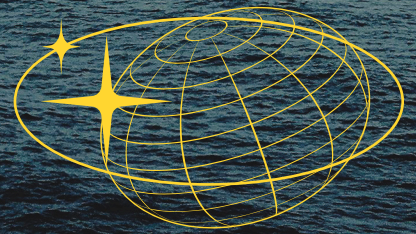


Quantum Optimization for Global Disaster Evacuation

Team Nexus



Problem

Natural disasters demand rapid and efficient evacuation strategies. Current systems often fall short due to:

- Poor planning leading to congestion and delays
- Increased risk to evacuees
- Inability to scale effectively with growing populations

Classical Limitation

Evacuation planning is fundamentally a complex combinatorial optimization problem. As the size of road networks and populations grows, classical algorithms become inefficient, hindering real-time decision-making.

Quantum Approach

Q-Evac leverages the power of quantum computing to overcome these limitations:

- Models the evacuation problem as a Quadratic Unconstrained Binary Optimization (QUBO) problem.
- **Inputs:** Road networks, population density, shelter capacity, and real-time traffic conditions.
- **Goal:** Minimize total evacuation time and reduce congestion hotspots.
- Utilizes quantum annealing to evaluate multiple evacuation solutions concurrently, identifying the optimal strategy.
- Designed for implementation on Infleqtion's neutral atom quantum processors.

Expected Impact

Q-Evac promises to deliver significant improvements in disaster response:

- **Faster Evacuation:** Optimized routes and reduced congestion lead to quicker evacuation times.
- **Reduced Congestion:** Real-time adjustments minimize bottlenecks and improve traffic flow.
- **Improved Emergency Response:** Enhanced decision-making capabilities for emergency responders.
- **Global Applicability:** Scalable solution applicable to diverse geographic locations and disaster scenarios.

Team Nexus

A team of five dedicated undergraduate students applying emerging quantum technology to solve critical real-world problems in disaster management.